

NEXFOLIO: COMPREHENSIVE PROJECT STRATEGIC ROADMAP

AI-POWERED INVESTMENT INTELLIGENCE PLATFORM

Project Designation: NexFolio – An Advanced AI-Driven Investment Intelligence & Risk Analytics Ecosystem

Academic Context: B.Tech Computer Science & Engineering Major Capstone Project

Frontend Architecture: Next.js 15 Enterprise Framework, React.js, Tailwind CSS, High-Fidelity Data Visualization via Recharts/Tremor

Asynchronous Backend: Python-based FastAPI for High-Performance Concurrency

Data Persistence: Scalable NoSQL Architecture via MongoDB & MongoDB Atlas Cloud

Security & Identity: Firebase Authentication Infrastructure with Integrated Google OAuth 2.0 Protocols

Artificial Intelligence Core: Scikit-learn, Gradient Boosted Decision Trees (XGBoost), SHAP Explainability Framework, Deep Learning via TensorFlow/Keras

Computational Analysis: Vectorized Data Processing using Pandas and NumPy

Market Intelligence Integration: Real-time Financial Aggregation via Yahoo Finance and Alpha Vantage APIs

Development Lifecycle: Distributed Version Control via Git & GitHub Architecture

Production Deployment: Edge-optimized Hosting on Vercel (Frontend) and Resilient Cloud Provisioning via Render/Railway (Backend)

MASTER PHASE MAP

Phase ID	Phase Name/Description
Phase 0	PROJECT PLANNING & ARCHITECTURE
Phase 1	DATABASE & SYSTEM DOCUMENTATION
Phase 2	AUTHENTICATION & USER MANAGEMENT
Phase 3	PORTFOLIO MANAGEMENT SYSTEM
Phase 4	MARKET DATA & DATA ENGINEERING
Phase 5	FEATURE ENGINEERING

Phase ID	Phase Name/Description
Phase 6	DATASET PREPARATION & MODEL FOUNDATION
Phase 7	AI/ML RISK INTELLIGENCE & EXPLAINABILITY
Phase 8	NEXT.JS FRONTEND INTEGRATION
Phase 9	INVESTMENT DASHBOARD & VISUALIZATION
Phase 10	AI RECOMMENDATION & PORTFOLIO INTELLIGENCE
Phase 11	LIVE MARKET DATA & ADVANCED FEATURES
Phase 12	SECURITY, VALIDATION & ERROR HANDLING
Phase 13	TESTING & QUALITY ASSURANCE
Phase 14	DEPLOYMENT & PRODUCTION
Phase 15	DOCUMENTATION & ACADEMIC REPORT
Phase 16	FINAL PRESENTATION & VIVA
Phase 17	FINAL RELEASE & GITHUB PORTFOLIO

PHASE 0 – PROJECT PLANNING & ARCHITECTURE

0.1 Project Identification

- Project title
- Problem statement
- Motivation
- Objectives
- Scope
- Target users

0.2 Requirement Analysis

- Functional requirements
- Non-functional requirements
- User requirements
- System requirements

0.3 Technology Selection

- Next.js
- FastAPI
- MongoDB
- Firebase
- Python ML stack
- Market-data APIs

0.4 System Architecture

- Frontend architecture
- Backend architecture
- AI/ML architecture
- Database architecture
- Authentication architecture
- API communication

0.5 Project Repository

- GitHub repository
- Root folder structure
- Branch strategy
- `.gitignore`
- README

STATUS:  COMPLETED

PHASE 1 – DATABASE & SYSTEM DOCUMENTATION

1.1 MongoDB Architecture

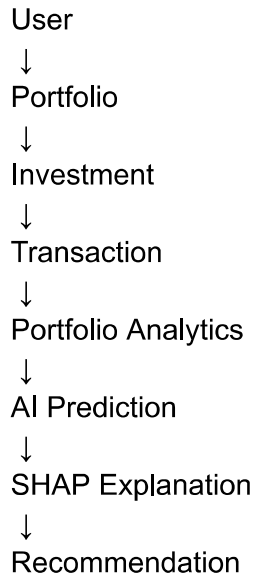
Design collections for:

- users
- portfolios
- investments
- transactions
- predictions
- explanations
- recommendations
- watchlist

- market data

1.2 Data Relationships

Define logical relationships between:



1.3 Documentation

- SRS
- ER/Data relationship documentation
- Architecture diagram
- Data-flow diagram
- Use-case diagram
- API documentation

1.4 Database Security

- User ownership
- Authentication-based access
- Validation
- No unauthorized portfolio access

STATUS:  **MOSTLY COMPLETED**

PHASE 2 – AUTHENTICATION & USER MANAGEMENT

2.1 Firebase Authentication

Implementation requirements include:

- Firebase project
- Google Authentication
- Firebase configuration
- Google Sign-In
- Authentication state

2.2 Frontend Authentication

- Login page
- Google Sign-In button
- Logout
- Auth state provider
- Protected routes

2.3 FastAPI Authentication

- Firebase Admin SDK
- ID-token verification
- Authentication middleware/dependency
- User identity extraction

2.4 User Management

Data persistence includes:

- Firebase UID
- Name
- Email
- Profile information
- Created timestamp

2.5 Authorization

Security protocols ensure strict isolation:

User A



Only User A's portfolios

User B



Only User B's portfolios

STATUS: ● **PARTIALLY PLANNED**

PHASE 3 – PORTFOLIO MANAGEMENT SYSTEM

3.1 Portfolio Creation

- Create portfolio
- Update portfolio
- Delete portfolio
- Portfolio ownership

3.2 Investment Management

- Add investment
- Update investment
- Remove investment
- Quantity
- Buy price
- Current price
- Investment value

3.3 Transaction Management

Standard operations support:

- BUY
- SELL

Schema requirements:

- symbol
- quantity
- price
- transaction date
- transaction type

3.4 Portfolio Analytics

Analytical metrics:

- Total investment
- Current value
- Profit/Loss
- ROI
- Asset allocation
- Sector allocation
- Number of assets
- Portfolio returns

3.5 Backend API

Defined API endpoints:

POST /portfolios

GET /portfolios

GET /portfolios/{id}

PUT /portfolios/{id}

DELETE /portfolios/{id}

POST /investments

GET /investments

PUT /investments/{id}

DELETE /investments/{id}

POST /transactions

GET /transactions

STATUS:  **CORE LOGIC COMPLETED / INTEGRATION CONTINUES**

PHASE 4 – MARKET DATA & DATA ENGINEERING

4.1 Market Data Collection

Data points collected:

- Open
- High
- Low

- Close
- Adjusted Close
- Volume

4.2 Market Instruments

Asset classes supported:

- NSE equities
- ETFs
- Indices
- Crypto where applicable

4.3 Data Sources

Primary data source:

- Yahoo Finance

Secondary and future sources:

- Alpha Vantage

4.4 Data Cleaning

Preprocessing protocols:

- Missing values
- Duplicate records
- Invalid values
- Date normalization
- Outliers

4.5 Data Storage

Datasets for distinct pipeline stages:

- raw market data
- cleaned data
- portfolio data
- engineered features

STATUS:  **COMPLETED**

PHASE 5 – FEATURE ENGINEERING

5.1 Return Features

Engineered return metrics:

- return_1M
- return_3M
- return_6M
- return_1Y
- annualized_return

5.2 Volatility Features

Volatility analysis features:

- annualized_volatility
- downside_deviation_annualized
- rolling_volatility

5.3 Risk Features

Risk assessment features:

- portfolio_beta
- portfolio_max_drawdown
- rolling_max_drawdown_30d

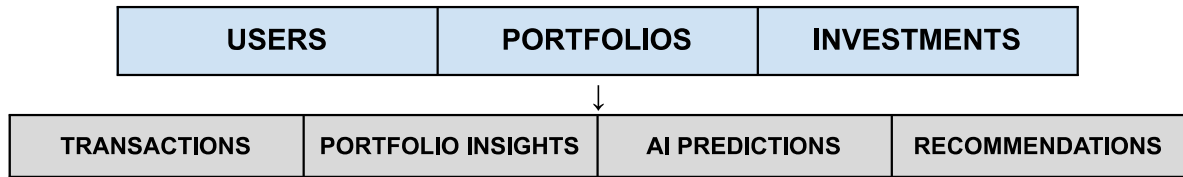
5.4 Performance Features

Performance evaluation features:

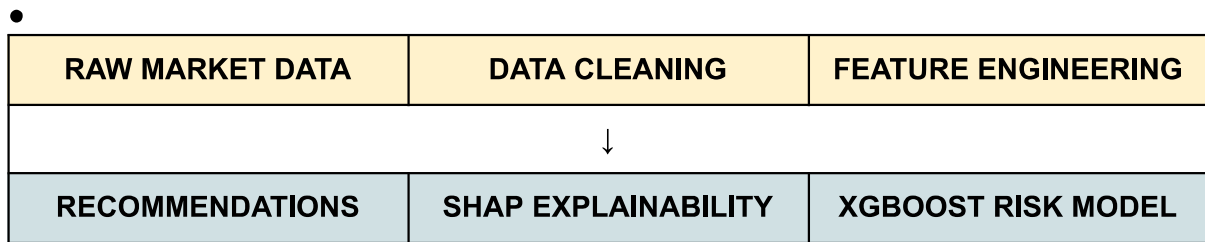
SYSTEM ARCHITECTURE DIAGRAM

USER AUTH (Firebase)	FRONTEND (Next.js)	BACKEND (FastAPI)
↓		↓
AI/ML ENGINE (XGBoost / SHAP)		DATABASE (MongoDB Atlas)

DATABASE ENTITY RELATIONSHIP DIAGRAM (ERD)



AI/ML PIPELINE FLOWCHART



- portfolio_sharpe_ratio
- portfolio_sortino_ratio
- portfolio_calmar_ratio

5.5 Portfolio Structure Features

Structural analysis features:

- asset_count
- sector_count
- sector allocation
- diversification score

5.6 Feature Store

Data lifecycle:

raw data
↓
cleaned data
↓
feature engineering
↓
feature dataset
↓
ML-ready dataset

PHASE 6 – DATASET PREPARATION & MODEL FOUNDATION

6.1 Dataset Audit

- Dataset validation
- Missing-value analysis
- Distribution analysis
- Feature inspection

6.2 Target Definition

Risk classification tiers:

LOW
MEDIUM
HIGH

6.3 Leakage Detection

Exclusion of leakage-sensitive features for model integrity.

Phase 7 refinement excluded the following:

hhi
diversification_score
largest_sector_pct
top_3_holdings_pct
top_5_holdings_pct
concentration_warning
volatility_warning
diversification_warning

6.4 Dataset Split

Current dataset parameters:

Total samples: 1000

Features: 36

Training: 800

Testing: 200

6.5 ML Artifacts

Exported artifacts:

X_train.parquet

X_test.parquet

y_train.parquet

y_test.parquet

feature_metadata.json

STATUS:  **COMPLETED**

PHASE 7 – AI/ML RISK INTELLIGENCE

This section represents the **primary intelligence core** of NexFolio.

7.1 Baseline Models

Benchmarking models include:

- Logistic Regression
- Decision Tree
- Random Forest

Experimental objectives:

- Establish benchmark
- Compare models
- Demonstrate model selection

STATUS:  **COMPLETED**

7.2 Random Forest Risk Model

Model training:

RandomForestClassifier

Categorical risk classes:

LOW

MEDIUM

HIGH

Evaluation metrics:

- Accuracy
- Precision
- Recall
- F1
- Confusion Matrix

STATUS:  **COMPLETED**

7.3 Model Benchmarking

Hierarchical model comparison:

Logistic Regression



Decision Tree



Random Forest



XGBoost

Performance metrics:

- Accuracy
- Precision
- Recall
- F1-score

Select production model.

STATUS:  COMPLETED

7.4 XGBoost + Explainable AI

7.4.1 Dataset Preparation

1000 samples
36 features
800 training
200 testing

STATUS:  COMPLETED

7.4.2 XGBoost Model

Production-grade model:

XGBClassifier

Validated performance results:

Accuracy: 90.00%
Precision: 90.41%
Recall 90.00%
F1 Score: 89.98%

STATUS:  COMPLETED

7.4.3 Global SHAP Analysis

Determine which features influence risk predictions globally.

Principal feature drivers:

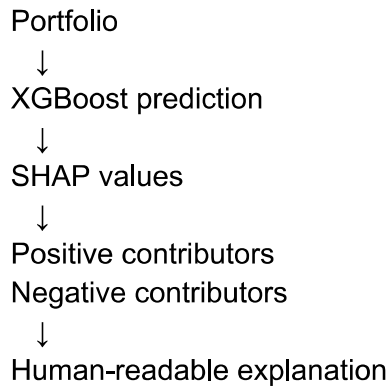
annualized_volatility
portfolio_beta
downside_deviation_annualized
asset_count
portfolio_sharpe_ratio

STATUS:  COMPLETED

7.4.4 Local SHAP Explanations

Explain individual portfolio predictions.

Sample explanation flow:



STATUS:  COMPLETED

7.4.5 Recommendation Engine

Recommendation logic based on:

- Risk category
- Volatility
- Diversification
- Asset count
- Sector concentration
- Drawdown
- Performance

STATUS:  COMPLETED

7.4.6 Portfolio Intelligence Report

Comprehensive investor reporting features:

- Risk category
- Annualized return
- Volatility
- Beta
- Diversification
- Asset count
- AI recommendations

STATUS:  COMPLETED

7.5 FASTAPI AI SERVICE

7.5.1 FastAPI Application

Directory structure:

```
app/
├── api/
├── services/
├── schemas/
├── config/
├── database/
└── main.py
```

7.5.2 Risk Prediction API

POST /api/v1/predict-risk

API response parameters:

risk_category
confidence
probabilities

STATUS:  COMPLETED

7.5.3 Explainability API

POST /api/v1/explain-risk

Return schema:

risk_category
confidence
top_positive_contributors
top_negative_contributors

STATUS:  COMPLETED

7.5.4 MongoDB Persistence

Store prediction results in MongoDB.

Persistence response schema:

prediction_id
risk_category
confidence

STATUS:  COMPLETED

7.5.5 Prediction History

History endpoints:

GET /api/v1/predictions
GET /api/v1/predictions/{prediction_id}

Functional scope:

- Prediction history
- Previous explanations
- User-specific AI history

STATUS:  NEXT BACKEND TASK / VERIFY CURRENT IMPLEMENTATION

PHASE 8 – NEXT.JS FRONTEND INTEGRATION

This phase represents the **immediate development priority**.

8.1 Frontend Environment

Environment variables and services:

NEXT_PUBLIC_API_BASE_URL
Firebase configuration

Connect Next.js → FastAPI.

STATUS: 🕒 NEXT

8.2 API Client

Create centralized API layer.

Service responsibilities:

- HTTP requests
- Authentication headers
- Error handling
- Response typing

Example architecture:

```
graph TD
    Nextjs[Next.js] --> APIClient[API Client]
    APIClient --> FastAPI[FastAPI]
    FastAPI --> MLService[ML Service]
    MLService --> MongoDB[MongoDB]
```

8.3 Risk Prediction Interface

User interface components:

- Portfolio input form
- Risk prediction button
- Loading state
- Error state
- Result card

Data presentation:

Risk Category

Confidence
LOW probability
MEDIUM probability
HIGH probability

8.4 Explainability Interface

Build SHAP visualization.

Visualization components:

Why is my portfolio HIGH risk?

Positive Contributors



Negative Contributors



Technical library stack:

- Recharts
 - Horizontal bar charts
 - Cards
 - Tooltips
-

8.5 Prediction History

Historical data views:

- Prediction date
- Portfolio
- Risk
- Confidence

User interaction flow:

Click prediction



Prediction details



SHAP explanation



Recommendations

8.6 Firebase Google Authentication

Authentication pipeline:

Google Login



Firebase



Firebase ID Token



FastAPI



Token verification



MongoDB user

8.7 Protected Dashboard

Public state (Unauthenticated):

Login

Private state (Authenticated):

Dashboard

Portfolio

Predictions

Recommendations

Profile

STATUS: 🕒 UPCOMING

PHASE 9 – INVESTMENT DASHBOARD

9.1 Main Dashboard

Primary KPIs displayed:

- Total portfolio value
- Investment amount
- Profit/Loss
- ROI
- Risk category
- AI confidence

9.2 Portfolio Charts

Analytical visualizations:

- Portfolio value chart
- Profit/Loss chart
- Asset allocation
- Sector allocation

9.3 Risk Dashboard

Risk-specific metrics:

Risk Level
Confidence
Volatility
Beta
Sharpe Ratio
Sortino Ratio
Maximum Drawdown
Diversification

9.4 AI Insights

Dashboard section:

AI Portfolio Insight

Heuristic insights:

- Risk warning
- Diversification suggestion
- Volatility warning
- Defensive allocation suggestion

STATUS: 🕒 **UPCOMING**

PHASE 10 – AI RECOMMENDATION & PORTFOLIO INTELLIGENCE

10.1 Recommendation Engine Integration

Frontend service consumption:

/api/v1/recommend

10.2 Personalized Recommendations

Strategic recommendation categories:

- diversification suggestions
- risk reduction
- sector balancing
- defensive allocation
- concentration warnings

10.3 Investor Intelligence

Intel components:

Portfolio Health

Risk Assessment

Performance Analysis

AI Explanation

Recommendations

10.4 Investor Report

Downloadable export contents:

- Portfolio summary
- Risk assessment
- AI explanation
- Recommendations
- Charts

STATUS: ● ENGINE COMPLETED / UI PENDING

PHASE 11 – LIVE MARKET DATA & ADVANCED FEATURES

11.1 Live Prices

Integrate market data API.

11.2 Watchlist

Functional requirements:

- Add stocks
- Remove stocks
- View current price
- View change

11.3 Market Overview

Market indicators:

- NIFTY
- SENSEX
- Major stocks
- Market movement

11.4 WebSocket

Future enhancement:

Market API



WebSocket



FastAPI
↓
Next.js
↓
Live dashboard

11.5 News Integration

Contextual news feed:

- Company news
- Market news
- Portfolio-related news

STATUS: 🕒 FUTURE PHASE

PHASE 12 – SECURITY & VALIDATION

12.1 Authentication Security

- Firebase token validation
- Expired token handling
- Unauthorized request handling

12.2 API Security

- Request validation
- Rate limiting where appropriate
- CORS
- Secure environment variables

12.3 Database Security

Strict enforcement of ownership:

User ID
↓
Portfolio ownership
↓
Prediction ownership

12.4 Input Validation

Data validation constraints:

- quantities
- prices
- percentages
- symbols
- portfolio IDs
- prediction IDs

12.5 Error Handling

Standardized HTTP status codes:

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

422 Validation Error

500 Internal Server Error

STATUS: 🕒 UPCOMING

PHASE 13 – TESTING & QUALITY ASSURANCE

13.1 Backend Testing

Core service validation:

- API endpoints
- Authentication
- Database operations
- ML predictions
- SHAP explanations

13.2 Frontend Testing

UI/UX functional validation:

- Login
- Dashboard
- Forms
- Charts
- API errors
- Loading states

13.3 ML Testing

Model consistency checks:

- Model loading
- Feature ordering
- Prediction consistency
- Probability output
- SHAP output

13.4 Integration Testing

End-to-end integration flow:

Google Login



Next.js



FastAPI



XGBoost



SHAP



MongoDB



Next.js Dashboard

13.5 Performance Testing

Performance benchmarks:

- API response time
- Prediction latency
- Dashboard loading
- Database query performance

PHASE 14 – DEPLOYMENT

14.1 Frontend Deployment

Frontend production environment:

Vercel

14.2 Backend Deployment

Backend production environment:

Render / Railway

14.3 Database

Cloud data persistence:

MongoDB Atlas

14.4 Firebase

Security configuration:

- Production domain
- Authorized domains
- Google OAuth

14.5 Environment Variables

Security best practices (Environment Secrets):

API keys

Firebase secrets

MongoDB credentials

Service account credentials

14.6 Production CORS

Cross-Origin Resource Sharing (CORS) flow:

Vercel frontend



FastAPI backend

14.7 Production Testing

Production verification checklist:

- Login
- Portfolio operations
- Prediction
- Explainability
- Recommendations
- Database persistence

STATUS: 🕒 **UPCOMING**

PHASE 15 – PROJECT DOCUMENTATION

This phase is critical for the **CBIT Final-Year B.Tech CSE** submission requirements.

15.1 Technical Documentation

Architecture and system documentation:

- Architecture
- APIs
- Database
- ML pipeline
- Frontend
- Authentication

15.2 ML Documentation

AI pipeline documentation:

- Dataset
- Features

- Target variable
- Data preprocessing
- Train/test split
- Algorithms
- Model comparison
- XGBoost
- SHAP
- Evaluation metrics

15.3 Results

Performance evaluation results:

- Accuracy
- Precision
- Recall
- F1
- Confusion matrix
- Feature importance
- SHAP analysis

15.4 Screenshots

Visual evidence of system functionality:

- Login
- Dashboard
- Portfolio
- Risk prediction
- SHAP explanation
- Recommendations
- Prediction history

15.5 Academic Report

Academic report structure:

1. Abstract
2. Introduction
3. Problem Statement
4. Literature Survey
5. Existing System
6. Proposed System
7. Requirements
8. System Architecture
9. Database Design

10. Implementation
11. Machine Learning Methodology
12. Results
13. Testing
14. Conclusion
15. Future Scope
16. References

STATUS: ● DOCUMENTATION ALREADY STARTED / FINALIZATION PENDING

PHASE 16 – FINAL PRESENTATION & VIVA

16.1 Presentation

Deck components:

- Problem
- Motivation
- Objectives
- Architecture
- Technology stack
- ML pipeline
- Results
- Screenshots
- Future scope

16.2 ML Viva Preparation

Defense of methodology and theory:

- Why XGBoost?
- Why Random Forest?
- Why SHAP?
- What is data leakage?
- Why remove leakage-sensitive features?
- What is precision?
- What is recall?
- What is F1?
- What is a confusion matrix?
- What does 90% accuracy mean?
- Why train/test split?

- How does prediction work?

16.3 System Viva

Defense of architecture and tech stack:

- Next.js
- FastAPI
- MongoDB
- Firebase
- REST APIs
- Authentication flow
- API architecture
- Database design

16.4 Demonstration

Live system walk-through:

Google Login



Dashboard



Create Portfolio



Add Investments



Portfolio Analytics



AI Risk Prediction



XGBoost



SHAP Explanation



AI Recommendations



Save Prediction



Prediction History

STATUS: 🕒 **FINAL STAGE**

PHASE 17 – FINAL RELEASE & GITHUB

17.1 Repository Cleanup

Sanitization of the production repository:

- unnecessary files
- temporary datasets
- credentials
- cache
- generated junk

17.2 GitHub Structure

Canonical folder structure:

```
NexFolio/  
|  
├── frontend/  
├── ai-service/  
├── database/  
├── docs/  
├── postman/  
├── README.md  
└── .gitignore
```

17.3 README

Project documentation components:

- Project overview
- Features
- Architecture
- Tech stack
- Installation
- API documentation
- ML results
- Screenshots
- Deployment
- Future scope

17.4 Release

Production release versioning:

17.5 Final GitHub Portfolio

Highlighted competencies:

- Architecture
- Real ML training
- XGBoost
- SHAP
- FastAPI
- MongoDB
- Firebase
- Next.js
- Deployment

STATUS: ● INITIAL GITHUB PUSH COMPLETED

THE BIG-PICTURE NEXFOLIO PIPELINE

